Name:……**Centre/Index number**……/……

Signature:……

UCE

553/2

BIOLOGY

PRACTICAL

2 Hours

Uganda Certificate of Education

INTERNAL MOCK EXAMINATIONS

BIOLOGY PRACTICAL

Paper 2

Time allowed: 2Hours

INSTRUCTION TO CANDIDATES

Answer all questions.

Answers should be written in the spaces provided.

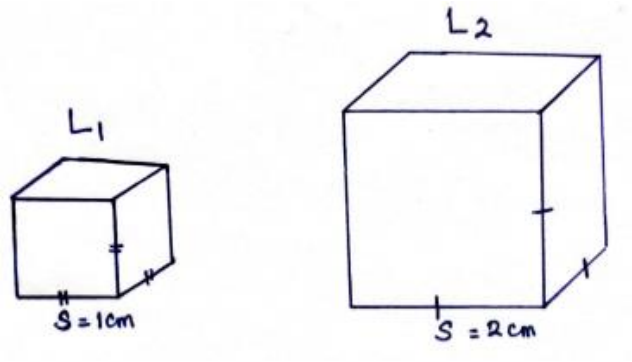
Use a sharp pencil for any drawing.

Additional answer sheets should not be inserted.

Answers written in additional sheet of paper will not be marked.

For examiners' use only		
Section/Question	**Marks**	**Examiner's signature**
1		
2		
3		
Total		

1. You are provided with specimen **L** and a solution of potassium permanganate. Using a knife, peel specimen **L**, then cut out two **cubes** from it, **L1** and **L2** each with side length **S**, 1 cm and 2cm respectively.as shown in the figure below.

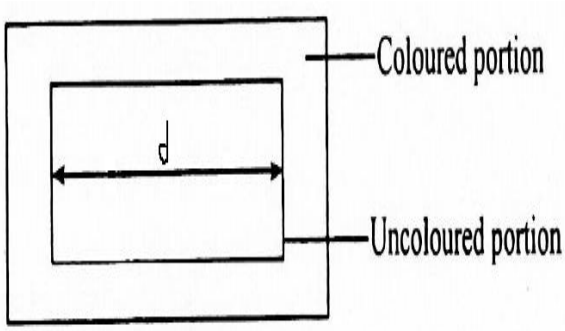


a. Calculate the surface area, volume and surface area to volume ratio of each cube and complete the table below; (6marks)

Cube	Surface areas (cm^2)	Volume (cm^3)	Surface area: volume ratio (cm^{-1})
L1			
L2			

b. Fill a clean beaker with $100cm^3$ of potassium permanganate solution. Then immerse or dip **L1** and **L2** in a beaker containing potassium permanganate solution to completely cover the cubes. Leave the setup to stand for 30minutes, after 30minutes, remove **L1** and **L2** and wipe them using a blotting/filter paper to remove the drops of potassium permanganate.

Using a razorblade, cut each cube into two halves, using one half of each cube, measure the distance(**d**) in mm across the uncoloured position as indicated in the figure below.



i. Record your results and complete the table below (6 marks)

Cube	Un coloured length (d) mm	Total colored length (S-d) mm	Percentage penetration $\frac{(S-d)\times 100}{S}$
L1			
L2			

ii. Explain how the results in b (i) above related to the physiological process named in b (ii), in living organisms. (4 marks)

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iii. Name the physiological process being investigated (1mark)

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c. State three importance of the process named in b (ii) to plants and animals.(3 marks)

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2. You are provided with specimens labelled **M** and **N** which are plant structures.

a. Identify the specimen and state two reasons for the identification.

i. **M** (1 mark)

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Reasons (2 marks)

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ii. **N** (1 mark)

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Reasons (2 marks)

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b. Cut specimen **M** longitudinally and state any four similarities between specimen **M** and **N**. (03 marks)

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c. Describe how each specimen is adapted to its function(s).

i. M (02 marks)

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ii. N (02 marks)

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d. Draw a longitudinal section of specimen **M.** (06 marks)

e. State two functions of specimen M to the plant. (1 mark)

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3. You are provided with specimens **A**, **B** and **C** which are animals belonging to the same class but living in different habitats.

a. Giving two reasons, identify the class to which they belong.

Class (1 mark)

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Reasons (2 marks)

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b. Examine the outer and inner wings of specimen **B**, and state any three structural differences between them. (3 marks)

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c. i) Examine the features on the head and the hind leg of each of the specimen. (06 marks)

Specimen	**Head**	**Hind leg**
A		
B		
C		

ii) Construct a dichotomous key using the structural features on the head described in (c)(i) above. (03 marks)

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d. Carefully remove the hind limb of specimen **A** from its body. Examine it using a hand lens and draw but don't label. (05 marks)

--(THE END)--